Chapter 7. Logit Models for Multinomial Responses

Baseline-category and cumulative logits

Contents

7.1 Nominal responses: baseline-category logit models **2**
- 7.1.1 Baseline-category logits . . . 2
- 7.1.2 Alligator food choice example . . . 2
- 7.1.3 Estimating response probabilities . . . 3
- 7.1.4 Fitting of baseline-category logit models* . . . 4
- 7.1.5 Multicategory logit model as multivariate GLM* . . . 5

7.2 Ordinal responses: cumulative logit models **5**
- 7.2.1 Cumulative logits . . . 5
- 7.2.2 Proportional odds model . . . 6
- 7.2.3 Latent variable motivation* . . . 7
- 7.2.4 Mental impairment example . . . 8
- 7.2.5 More complex models . . . 9

7.3 Ordinal responses: cumulative link models **9**
- 7.3.1 Types of cumulative links . . . 9
- 7.3.2 Estimation for cumulative link models . . . 10
- 7.3.3 Life table example . . . 10
- 7.3.4 Incorporating dispersion effects* . . . 10

7.4 Alternative models for ordinal responses* **11**
- 7.4.1 Adjacent-categories logits . . . 11
- 7.4.2 Job satisfaction example . . . 12
- 7.4.3 Continuation-ratio logits . . . 12
- 7.4.4 Developmental toxicity study with pregnant mice . . . 13
- 7.4.5 Mean response models for ordered response . . . 14
- 7.4.6 Job satisfaction example revisited . . . 14
- 7.4.7 Advantages and disadvantages of mean response models . . . 14

7.5 Testing conditional independence in $I \times J \times K$ tables* **14**
- 7.5.1 Using multinomial models to test conditional independence . . . 14
- 7.5.2 Job satisfaction example revisited . . . 15
- 7.5.3 Generalized Cochran–Mantel–Haenszel tests for $I \times J \times K$ tables . . . 15
- 7.5.4 Job satisfaction example revisited . . . 16
- 7.5.5 Related score tests for multinomial logit models . . . 16
- 7.5.6 Exact tests of conditional independence . . . 17

7.6 Discrete-choice multinomial logit models* **17**
- 7.6.1 Discrete-choice modeling . . . 18
- 7.6.2 Discrete-choice and multinomial logit models . . . 18
- 7.6.3 Shopping choice example . . . 19

7.7 R lab **19**

7.1 Nominal responses: baseline-category logit models

Motivation. Many responses have more than two unordered categories (food type, brand, party). Pairing each category with a baseline gives $J-1$ simultaneous logit equations that together describe all pairwise odds, extending binary logistic regression.

Let Y have J categories and $\pi_j(\mathbf{x}) = P(Y = j \mid \mathbf{x})$, $\sum_j \pi_j(\mathbf{x}) = 1$. At a fixed setting $\mathbf{x}$, the counts in the J categories are multinomial with probabilities $\{\pi_1(\mathbf{x}), \ldots, \pi_J(\mathbf{x})\}$. Of the $\binom{J}{2}$ pairwise logits, any suitable $J-1$ determine the rest.

7.1.1 Baseline-category logits

Definition (Baseline-category logit model)

With category J as baseline,

$$\log \frac{\pi_j(\mathbf{x})}{\pi_J(\mathbf{x})} = \alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x}, \qquad j = 1, \ldots, J-1. \tag{7.1}$$

The effects $\boldsymbol{\beta}_j$ depend on which category is paired with the baseline.

Proposition 7.1 (All pairwise logits)

Under (7.1), for any categories a, b,

$$\log \frac{\pi_a(\mathbf{x})}{\pi_b(\mathbf{x})} = (\alpha_a - \alpha_b) + (\boldsymbol{\beta}_a - \boldsymbol{\beta}_b)^\top \mathbf{x},$$

with the convention $\alpha_J = 0$, $\boldsymbol{\beta}_J = \mathbf{0}$. Hence the choice of baseline is arbitrary: changing it reparametrizes but does not change the model or its fit.

Proof. $\log(\pi_a/\pi_b) = \log(\pi_a/\pi_J) - \log(\pi_b/\pi_J)$; substitute (7.1) (the convention makes the formula valid when a or b equals J). The set of probability vectors described is the same for every baseline, since each parametrization is a linear reparametrization of the others. ■

With categorical predictors and nonsparse data, X^2 and G^2 check fit; with continuous predictors or sparse data they remain valid for comparing nested models differing by few terms.

7.1.2 Alligator food choice example

Example 7.2 (Alligator food choice)

For 219 alligators from four Florida lakes, the response is the primary food type by volume (fish, invertebrate, reptile, bird, other); predictors are lake L, gender G and size S (≤ 2.3 m vs. > 2.3 m). The data are sparse (219 observations in 80 cells), so G^2 is more reliable for comparing models than for testing fit. Comparisons give $G^2[(G) \mid (\cdot)] = 2.1$ and $G^2[(L+S) \mid (G+L+S)] = 2.2$, each on $df = 4$, so we collapse over gender. For the collapsed table, $(L+S)$ has $G^2 = 17.1$, $X^2 = 15.0$ ($df = 12$), whereas (L) has 38.2 and (S) 66.2; both L and S matter. Observed counts (fitted values for $L+S$ in parentheses):

Lake	Size	Fish	Invert.	Reptile	Bird	Other
Hancock	≤ 2.3	23 (20.9)	4 (3.6)	2 (1.9)	2 (2.7)	8 (9.9)
	> 2.3	7 (9.1)	0 (0.4)	1 (1.1)	3 (2.3)	5 (3.1)
Oklawaha	≤ 2.3	5 (5.2)	11 (12.0)	1 (1.5)	0 (0.2)	3 (1.1)
	> 2.3	13 (12.8)	8 (7.0)	6 (5.5)	1 (0.8)	0 (1.9)
Trafford	≤ 2.3	5 (4.4)	11 (12.4)	2 (2.1)	1 (0.9)	5 (4.2)
	> 2.3	8 (8.6)	7 (5.6)	6 (5.9)	3 (3.1)	5 (5.8)
George	≤ 2.3	16 (18.5)	19 (16.9)	1 (0.5)	2 (1.2)	3 (3.8)
	> 2.3	17 (14.5)	1 (3.1)	0 (0.5)	1 (1.8)	3 (2.2)

Only two of the 40 standardized residuals exceed 2 in absolute value. With fish as baseline, dummies $s = 1$ for small size and z_H, z_O, z_T for Hancock, Oklawaha, Trafford (George is the reference), the ML estimates (SE) are:

Logit	Intercept	Size ≤ 2.3	Hancock	Oklawaha	Trafford
$\log(\pi_I/\pi_F)$	-1.55	1.46 (0.40)	-1.66 (0.61)	0.94 (0.47)	1.12 (0.49)
$\log(\pi_R/\pi_F)$	-3.31	-0.35 (0.58)	1.24 (1.19)	2.46 (1.12)	2.94 (1.12)
$\log(\pi_B/\pi_F)$	-2.09	-0.63 (0.64)	0.70 (0.78)	-0.65 (1.20)	1.09 (0.84)
$\log(\pi_O/\pi_F)$	-1.90	0.33 (0.45)	0.83 (0.56)	0.01 (0.78)	1.52 (0.62)

Given lake, small alligators have estimated odds of invertebrates rather than fish $e^{1.46} = 4.3$ times those of large alligators (Wald 95% CI $e^{1.46 \pm 1.96(0.396)} = (2.0, 9.3)$). Those odds are relatively higher in Trafford and Oklawaha and lower in Hancock than in George. Other pairs follow by subtraction (Proposition 7.1), e.g.

$$\begin{aligned}\log(\hat{\pi}_I/\hat{\pi}_O) &= (-1.55 + 1.46s - 1.66z_H + 0.94z_O + 1.12z_T) \\ &\quad - (-1.90 + 0.33s + 0.83z_H + 0.01z_O + 1.52z_T) \\ &= 0.35 + 1.13s - 2.48z_H + 0.93z_O - 0.39z_T.\end{aligned}$$

7.1.3 Estimating response probabilities

Proposition 7.3 (Response probabilities)

Model (7.1) is equivalent to

$$\pi_j(\mathbf{x}) = \frac{\exp(\alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x})}{1 + \sum_{h=1}^{J-1} \exp(\alpha_h + \boldsymbol{\beta}_h^\top \mathbf{x})}, \qquad j = 1, \ldots, J, \tag{7.2}$$

with $\alpha_J = 0$, $\boldsymbol{\beta}_J = \mathbf{0}$. For $J = 2$ this is binary logistic regression.

Proof. From (7.1), $\pi_j = \pi_J \exp(\eta_j)$ with $\eta_j = \alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x}$, valid also for $j = J$ with $\eta_J = 0$. Summing over j, $1 = \pi_J \sum_{h=1}^{J} e^{\eta_h} = \pi_J(1 + \sum_{h<J} e^{\eta_h})$. Solve for π_J and substitute. Conversely (7.2) gives $\pi_j/\pi_J = e^{\eta_j}$. ■

The parameters of the baseline category are set to zero for identifiability: adding a common vector to all $(\alpha_j, \boldsymbol{\beta}_j)$, $j = 1, \ldots, J$, leaves (7.2) unchanged.

Example 7.4 (Alligators: fitted probabilities)

For a large alligator in Lake Hancock ($s = 0$, $z_H = 1$),

$$\hat{\pi}_I = \frac{e^{-1.55-1.66}}{1 + e^{-1.55-1.66} + e^{-3.31+1.24} + e^{-2.09+0.70} + e^{-1.90+0.83}} = 0.023,$$

and similarly $\hat{\pi}_R = 0.072$, $\hat{\pi}_B = 0.141$, $\hat{\pi}_O = 0.194$, $\hat{\pi}_F = 0.570$. With a quantitative predictor (actual length), plotting $\hat{\pi}_j(x)$ against x is informative; with $J > 2$ a single $\hat{\pi}_j(x)$ need not be monotone in x.

7.1.4 Fitting of baseline-category logit models*

For subject i let $\mathbf{y}_i = (y_{i1}, \ldots, y_{iJ})$ indicate the response category ($\sum_j y_{ij} = 1$), $\mathbf{x}_i$ the predictors (with $x_{i0} = 1$), and $\boldsymbol{\beta}_j$ the parameters of the jth logit (including $\alpha_j = \beta_{j0}$).

Theorem 7.5 (Multinomial log likelihood)

(a) The baseline-category logits $\log(\pi_j/\pi_J)$ are the natural parameters of the multinomial distribution.
(b) For independent subjects, the log likelihood for (7.1) is

$$L(\boldsymbol{\beta}) = \sum_{j=1}^{J-1} \sum_{k} \beta_{jk} \Big(\sum_i x_{ik} y_{ij} \Big) - \sum_i \log\Big[1 + \sum_{j=1}^{J-1} \exp(\boldsymbol{\beta}_j^\top \mathbf{x}_i)\Big],$$

so the sufficient statistics are $\sum_i x_{ik} y_{ij}$; in particular $\sum_i y_{ij}$ (the count in category j) for α_j.
(c) The likelihood equations equate sufficient statistics to their expectations, $\sum_i x_{ik} y_{ij} = \sum_i x_{ik} \hat{\pi}_j(\mathbf{x}_i)$, and L is concave.

Proof.

Step 1. Using $\pi_J = 1 - \sum_{j<J} \pi_j$ and $y_{iJ} = 1 - \sum_{j<J} y_{ij}$, subject i contributes

$$\sum_{j=1}^{J} y_{ij} \log \pi_j(\mathbf{x}_i) = \sum_{j=1}^{J-1} y_{ij} \log \frac{\pi_j(\mathbf{x}_i)}{\pi_J(\mathbf{x}_i)} + \log \pi_J(\mathbf{x}_i),$$

an exponential-family form with natural parameters $\log(\pi_j/\pi_J)$, proving (a).

Step 2. Substitute $\log(\pi_j/\pi_J) = \boldsymbol{\beta}_j^\top \mathbf{x}_i$ and $\pi_J = [1 + \sum_{j<J} \exp(\boldsymbol{\beta}_j^\top \mathbf{x}_i)]^{-1}$ (Proposition 7.3) and sum over i; collecting the coefficient of β_{jk} gives (b).

Step 3. Differentiating, $\partial L/\partial \beta_{jk} = \sum_i x_{ik}[y_{ij} - \pi_j(\mathbf{x}_i)]$, which gives the likelihood equations. Differentiating again,

$$-\frac{\partial^2 L}{\partial \beta_{jk}\, \partial \beta_{hl}} = \sum_i x_{ik} x_{il}\, \pi_j(\mathbf{x}_i)[\delta_{jh} - \pi_h(\mathbf{x}_i)].$$

For each i, the $(J-1) \times (J-1)$ matrix $[\pi_j(\delta_{jh} - \pi_h)]$ is the covariance matrix of $(y_{i1}, \ldots, y_{i,J-1})$, hence positive semidefinite. The negative Hessian is $\sum_i \Sigma_i \otimes \mathbf{x}_i \mathbf{x}_i^\top$, a sum of Kronecker products of PSD matrices, so it is PSD and L is concave. ∎

Newton–Raphson thus converges to the ML estimates (when they exist); they are asymptotically normal with covariance the inverse information.

Proposition 7.6 (Separate binary fits)

The jth baseline-category logit is an ordinary logit of conditional probabilities:

$$\log \frac{\pi_j(\mathbf{x})}{\pi_J(\mathbf{x})} = \operatorname{logit} P(Y = j \mid Y \in \{j, J\}, \mathbf{x}).$$

Hence fitting a binary logistic regression to the observations in categories j and J alone consistently estimates $(\alpha_j, \boldsymbol{\beta}_j)$.

Proof. $P(Y = j \mid Y \in \{j, J\}) = \pi_j/(\pi_j + \pi_J)$, whose odds are π_j/π_J. Given membership in $\{j, J\}$, the responses are independent Bernoulli with this probability, so the binary fit is an ML fit of a correctly specified model. ■

Separate fits are less efficient than the simultaneous ML fit, but the loss is minor when the baseline is the most prevalent category. For the alligators, fitting invertebrate vs. fish alone gives $\log(\hat{\pi}_I/\hat{\pi}_F) = -1.69 + 1.66s - 1.78z_H + 1.05z_O + 1.22z_T$, with SEs $0.43, 0.62, 0.49, 0.52$ for the effects, close to the joint fit, since 155 of the 219 alligators chose fish or invertebrates.

7.1.5 Multicategory logit model as multivariate GLM*

A multivariate GLM for a vector response $\mathbf{y}_i$ with $\boldsymbol{\mu}_i = E(\mathbf{Y}_i)$ has the form $\mathbf{g}(\boldsymbol{\mu}_i) = X_i\boldsymbol{\beta}$, where row h of X_i holds the predictor values for component y_{ih}. The baseline-category logit model is of this form with $\mathbf{y}_i = (y_{i1}, \ldots, y_{i,J-1})$ (y_{iJ} is redundant), $g_j(\boldsymbol{\mu}_i) = \log[\mu_{ij}/(1 - \mu_{i1} - \cdots - \mu_{i,J-1})]$, and

$$X_i = \begin{pmatrix} 1 & \mathbf{x}_i^\top & & & & \\ & & 1 & \mathbf{x}_i^\top & & \\ & & & & \ddots & \\ & & & & 1 & \mathbf{x}_i^\top \end{pmatrix}, \qquad \boldsymbol{\beta} = (\alpha_1, \boldsymbol{\beta}_1^\top, \ldots, \alpha_{J-1}, \boldsymbol{\beta}_{J-1}^\top)^\top,$$

with zeros elsewhere. The same applies to grouped data using sample proportions.

7.2 Ordinal responses: cumulative logit models

Motivation. When the categories are ordered, a model that uses the ordering is more parsimonious and has more power (Section 6.4). Cumulative logits with a common effect across cutpoints give the most popular such model, motivated by an underlying continuous latent response.

7.2.1 Cumulative logits

Definition (Cumulative logits)

With $P(Y \le j \mid \mathbf{x}) = \pi_1(\mathbf{x}) + \cdots + \pi_j(\mathbf{x})$,

$$\operatorname{logit} P(Y \le j \mid \mathbf{x}) = \log \frac{P(Y \le j \mid \mathbf{x})}{1 - P(Y \le j \mid \mathbf{x})} = \log \frac{\pi_1(\mathbf{x}) + \cdots + \pi_j(\mathbf{x})}{\pi_{j+1}(\mathbf{x}) + \cdots + \pi_J(\mathbf{x})}, \quad j = 1, \ldots, J-1.$$

Each cumulative logit uses all J categories; a model for one of them alone is a binary logit model for the collapsing $\{1, \ldots, j\}$ vs. $\{j+1, \ldots, J\}$.

7.2.2 Proportional odds model

Definition (Proportional odds model)

$$\text{logit}\, P(Y \le j \mid \mathbf{x}) = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1, \tag{7.3}$$

with intercepts $\alpha_1 < \alpha_2 < \cdots < \alpha_{J-1}$ (since $P(Y \le j \mid \mathbf{x})$ increases in j) and a single effect vector $\boldsymbol{\beta}$ common to all cutpoints.

Proposition 7.7 (Parallel curves and cumulative odds ratios)

Under (7.3):
(a) for a single predictor and $\beta \neq 0$, the curve of $P(Y \le k \mid x)$ is the curve of $P(Y \le j \mid x)$ shifted horizontally: $P(Y \le k \mid X = x) = P(Y \le j \mid X = x + (\alpha_k - \alpha_j)/\beta)$;
(b) for any $\mathbf{x}_1, \mathbf{x}_2$ and every j, the cumulative odds ratio is

$$\frac{P(Y \le j \mid \mathbf{x}_1)/P(Y > j \mid \mathbf{x}_1)}{P(Y \le j \mid \mathbf{x}_2)/P(Y > j \mid \mathbf{x}_2)} = \exp[\boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)],$$

the same for all j and proportional (on the log scale) to the distance between $\mathbf{x}_1$ and $\mathbf{x}_2$;
(c) the conditional distributions are stochastically ordered: for each pair $\mathbf{x}_1, \mathbf{x}_2$, either $P(Y \le j \mid \mathbf{x}_1) \le P(Y \le j \mid \mathbf{x}_2)$ for all j, or $\ge$ for all j.

Proof. Let $F(z) = e^z/(1+e^z)$, so $P(Y \le j \mid \mathbf{x}) = F(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x})$.

(a) $F(\alpha_j + \beta[x + (\alpha_k - \alpha_j)/\beta]) = F(\alpha_k + \beta x)$.

(b) The log of the cumulative odds ratio is the difference of logits, $(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_1) - (\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_2) = \boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)$.

(c) F is increasing, so $P(Y \le j \mid \mathbf{x}_1) \le P(Y \le j \mid \mathbf{x}_2)$ iff $\boldsymbol{\beta}^\top \mathbf{x}_1 \le \boldsymbol{\beta}^\top \mathbf{x}_2$, a condition not involving j. ■

With one predictor, e^β is the cumulative odds ratio for $x_1 - x_2 = 1$, the same for every collapsing of Y into $(\le j, > j)$: hence McCullagh's name *proportional odds*. The response curves for $j = 1, 2, 3$ ($J = 4$) have exactly the same shape (Figure 1).

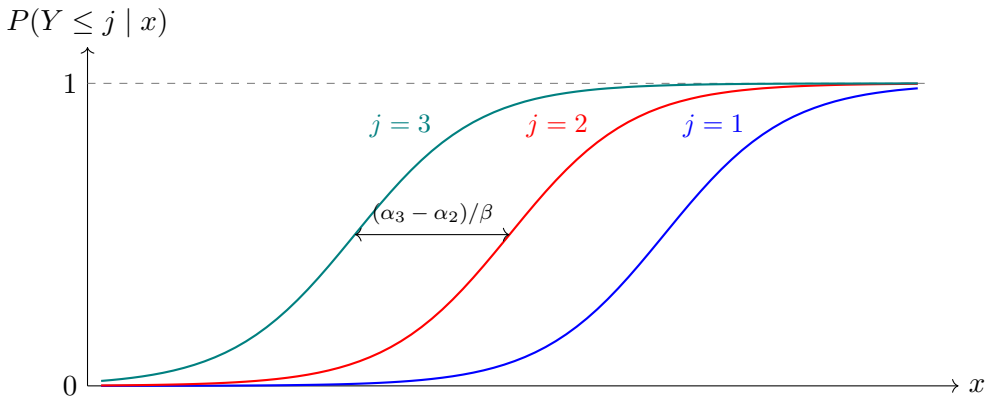


Figure 1: Proportional odds model with $J = 4$ and $\beta > 0$: the three cumulative probability curves are logistic curves with a common slope, horizontally displaced from one another.

Because the $J - 1$ curves are constrained to share a shape, the fit differs from separate binary fits.

With y_{ij} indicators of the response of subject i, the likelihood is

$$\prod_{i=1}^{n}\prod_{j=1}^{J}\Big[\frac{\exp(\alpha_j+\boldsymbol{\beta}^\top\mathbf{x}_i)}{1+\exp(\alpha_j+\boldsymbol{\beta}^\top\mathbf{x}_i)}-\frac{\exp(\alpha_{j-1}+\boldsymbol{\beta}^\top\mathbf{x}_i)}{1+\exp(\alpha_{j-1}+\boldsymbol{\beta}^\top\mathbf{x}_i)}\Big]^{y_{ij}} \tag{7.4}$$

with $\alpha_0=-\infty$, $\alpha_J=+\infty$; it is maximized by Fisher scoring.

7.2.3 Latent variable motivation*

Theorem 7.8 (Latent variable model implies a cumulative link model)

Let $Y^*=\boldsymbol{\beta}^\top\mathbf{x}+\varepsilon$, where ε has continuous cdf G, and let $-\infty=\alpha_0<\alpha_1<\cdots<\alpha_J=\infty$ be cutpoints with $Y=j$ iff $\alpha_{j-1}<Y^*\le\alpha_j$. Then

$$G^{-1}[P(Y\le j\mid\mathbf{x})]=\alpha_j-\boldsymbol{\beta}^\top\mathbf{x},\qquad j=1,\dots,J-1.$$

If G is the standard logistic cdf, this is the proportional odds model (with $-\boldsymbol{\beta}$ in place of $\boldsymbol{\beta}$); if $G=\Phi$, it is the cumulative probit model.

Proof. $P(Y\le j\mid\mathbf{x})=P(Y^*\le\alpha_j\mid\mathbf{x})=P(\varepsilon\le\alpha_j-\boldsymbol{\beta}^\top\mathbf{x})=G(\alpha_j-\boldsymbol{\beta}^\top\mathbf{x})$; apply G^{-1}. For the logistic cdf $G^{-1}=\text{logit}$. ■

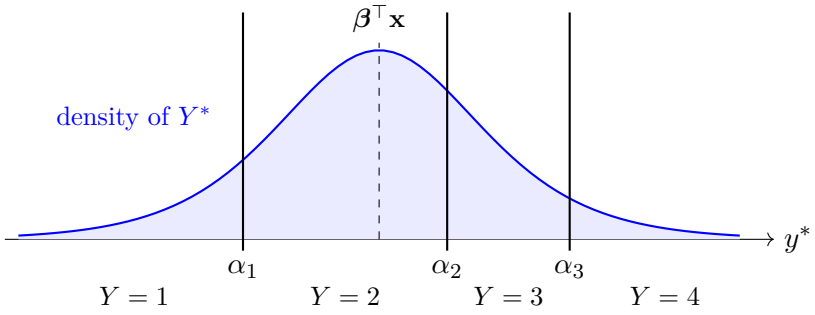


Figure 2: Latent-variable view of a cumulative link model ($J=4$): Y^* varies around $\boldsymbol{\beta}^\top\mathbf{x}$ and the observed category records the interval between cutpoints that contains it. Increasing $\boldsymbol{\beta}^\top\mathbf{x}$ shifts the density right and moves probability toward high categories.

Corollary 7.9 (Invariance to the choice of categories)

If (7.3) (or any cumulative link model) holds for Y, it holds with the same $\boldsymbol{\beta}$ for any variable obtained by merging adjacent categories of Y; the merged model simply omits the corresponding intercepts.

Proof. Merging categories j and $j+1$ removes the cutpoint between them: the cumulative probabilities of the new variable are a subset $\{P(Y\le j'\mid\mathbf{x}):j'\ne j\}$ of the old ones, each still equal to $G(\alpha_{j'}+\boldsymbol{\beta}^\top\mathbf{x})$. Equivalently, in Theorem 7.8 the latent regression is unchanged and only the cutpoints are coarsened. ■

Thus effects estimated with three categories (liberal, moderate, conservative) and with five (very liberal, …, very conservative) estimate the same parameters, making studies with different scales comparable. In the latent parametrization $\alpha_j-\boldsymbol{\beta}^\top\mathbf{x}$, $\beta>0$ means Y tends to be larger at higher x (usual sign meaning); much software (SAS) uses $\alpha_j+\boldsymbol{\beta}^\top\mathbf{x}$, whereas R's `MASS::polr` uses $\alpha_j-\boldsymbol{\beta}^\top\mathbf{x}$.

Proposition 7.10 (Reversing the category order)

Let $Y' = J + 1 - Y$. If G is symmetric ($G(-z) = 1 - G(z)$, e.g. logistic or normal) and $G^{-1}[P(Y \le j \mid \mathbf{x})] = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}$, then

$$G^{-1}[P(Y' \le j \mid \mathbf{x})] = -\alpha_{J-j} - \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1.$$

The model is the same model; only the signs of $\boldsymbol{\beta}$ and the (reordered) intercepts change. For an asymmetric G (complementary log-log), reversal produces the model with the reflected link (log-log).

Proof. $P(Y' \le j) = P(Y \ge J+1-j) = 1 - P(Y \le J-j) = 1 - G(\alpha_{J-j} + \boldsymbol{\beta}^\top \mathbf{x}) = G(-\alpha_{J-j} - \boldsymbol{\beta}^\top \mathbf{x})$ by symmetry. Apply G^{-1}. The new intercepts $-\alpha_{J-1} < \cdots < -\alpha_1$ are increasing in j, as required. Without symmetry, $1 - G(z) = \tilde{G}(-z)$ with $\tilde{G}(u) = 1 - G(-u)$, the reflected cdf; for $G(z) = 1 - \exp(-e^z)$ this gives $\tilde{G}(u) = \exp(-e^{-u})$, the log-log form. ■

7.2.4 Mental impairment example

Example 7.11 (Mental impairment)

A random sample of 40 adults in Alachua County, Florida. Y = mental impairment (1 well, 2 mild symptom formation, 3 moderate symptom formation, 4 impaired); x_1 = life events index (number and severity of important events in the past three years); x_2 = SES (1 high, 0 low). Listed as (x_2, x_1) within each response category:

Well	(1,1) (1,9) (1,4) (1,3) (0,2) (1,0) (0,1) (1,3) (1,3) (1,7) (0,1) (0,2)
Mild	(1,5) (0,6) (1,3) (0,1) (1,8) (1,2) (0,5) (1,5) (1,9) (0,3) (1,3) (1,1)
Moderate	(0,0) (1,4) (0,3) (0,9) (1,6) (0,4) (0,3)
Impaired	(1,8) (1,2) (1,7) (0,5) (0,4) (0,4) (1,8) (0,8) (0,9)

The fit of logit $P(Y \le j \mid \mathbf{x}) = \alpha_j + \beta_1 x_1 + \beta_2 x_2$:

Parameter	Estimate	SE	LR 95% CI
α_1	−0.282	0.642	(−1.56, 0.98)
α_2	1.213	0.661	(−0.05, 2.57)
α_3	2.209	0.721	(0.86, 3.71)
life (β_1)	−0.319	0.121	(−0.57, −0.09)
SES (β_2)	1.111	0.611	(−0.06, 2.35)

At the mean life events $\bar{x}_1 = 4.275$ and low SES,

$$\hat{P}(Y = 1) = \frac{\exp[-0.282 - 0.319(4.275)]}{1 + \exp[-0.282 - 0.319(4.275)]} = 0.16.$$

The cumulative probability from the "well" end decreases with life events and is higher at high SES: given life events, the odds of impairment below any fixed level at high SES are $e^{1.111} = 3.0$ times those at low SES. In probability terms, at $\bar{x}_1$, $\hat{P}(Y = 1)$ is 0.37 (high SES) vs. 0.16 (low SES); between the life-events quartiles 2.0 and 6.5 it falls from 0.55 to 0.22 (high SES) and from 0.28 to 0.09 (low SES). The score test of proportional odds (separate β's for each logit, 4 extra parameters) gives 2.33, $df = 4$, $P = 0.68$.

7.2.5 More complex models

Interactions, factors and so on enter as in binary logistic regression; only the intercept is replaced by $J-1$ cutpoint intercepts. For the mental impairment data, adding the interaction gives

$$\operatorname{logit} \hat{P}(Y \le j \mid \mathbf{x}) = \hat{\alpha}_j - 0.420x_1 + 0.371x_2 + 0.181x_1x_2,$$

interaction SE 0.238: the life-events effect is -0.420 at low SES and -0.239 at high SES, not significantly different.

When proportional odds fails

Replacing $\boldsymbol{\beta}$ by $\boldsymbol{\beta}_j$ gives nonparallel curves, which necessarily cross for some $\mathbf{x}$, violating the ordering of cumulative probabilities. A small P-value in the proportional odds test may reflect a large n more than important differences among the $\hat{\boldsymbol{\beta}}_j$, and the simpler model may have smaller MSE even if biased. If the fit is poor in practical terms, options include:
(1) a link with a nonsymmetric response curve (complementary log-log);
(2) extra terms such as interactions;
(3) dispersion parameters (Section 7.3.4);
(4) separate effects for some predictors only (partial proportional odds);
(5) a baseline-category logit model, using the ordering informally in interpretation.

7.3 Ordinal responses: cumulative link models

Motivation. The latent-variable argument works for any error distribution, so the logit can be replaced by other inverse cdfs. Probit and complementary log-log links give cumulative models with their own interpretations, and a scale component handles dispersion differences.

Definition (Cumulative link model)

For a continuous cdf G,

$$G^{-1}[P(Y \le j \mid \mathbf{x})] = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1. \tag{7.5}$$

By Theorem 7.8, (7.5) arises from crude measurement of a latent Y^* following a location family $G(y^* - \boldsymbol{\beta}^\top \mathbf{x})$; the α_j are cutpoints on a standardized latent scale (using $-\boldsymbol{\beta}$ instead of $+\boldsymbol{\beta}$ only flips signs).

7.3.1 Types of cumulative links

(a) *Cumulative probit* ($G = \Phi$): appropriate when Y^* is normal. In $\Phi^{-1}[P(Y \le j)] = \alpha_j - \boldsymbol{\beta}^\top \mathbf{x}$, a unit increase in x_k raises $E(Y^*)$ by β_k standard deviations of ε. Logit and probit fits are similar; logit parameters are easier to interpret.

(b) *Complementary log-log* ($G(z) = 1 - \exp(-e^z)$, from an extreme-value latent distribution): $\log\{-\log[1 - P(Y \le j \mid \mathbf{x})]\} = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}$. It is the grouped-time version of the proportional hazards model and is sometimes called a *proportional hazards* model. $P(Y \le j)$ approaches 1 faster than it approaches 0. The log-log link suits the case where cloglog holds for the reversed categories (Proposition 7.10).

Proposition 7.12 (Power interpretation of the cumulative cloglog model)

Under the cumulative complementary log-log model, for all j,

$$P(Y > j \mid \mathbf{x}_1) = P(Y > j \mid \mathbf{x}_2)^{\exp[\boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)]}.$$

Proof. $P(Y > j \mid \mathbf{x}) = 1 - G(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}) = \exp\left[-e^{\alpha_j} e^{\boldsymbol{\beta}^\top \mathbf{x}}\right]$. Hence

$$\log P(Y > j \mid \mathbf{x}_1) = -e^{\alpha_j} e^{\boldsymbol{\beta}^\top \mathbf{x}_2} e^{\boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)} = e^{\boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)} \log P(Y > j \mid \mathbf{x}_2);$$

exponentiate. ■

7.3.2 Estimation for cumulative link models

Cumulative link models are multivariate GLMs. McCullagh's Fisher scoring algorithm uses the likelihood written as in (7.4) with G in place of the logistic cdf; for large n the maximum is unique, and the log likelihood is concave for the logit, probit and complementary log-log links, so iterations converge rapidly.

7.3.3 Life table example

Example 7.13 (U.S. life lengths, 1981)

Percent distribution of life length by gender and race (fitted cloglog values in parentheses):

	Males		Females	
Life length	White	Black	White	Black
0–20	2.4 (2.4)	3.6 (4.4)	1.6 (1.2)	2.7 (2.3)
20–40	3.4 (3.5)	7.5 (6.4)	1.4 (1.9)	2.9 (3.4)
40–50	3.8 (4.4)	8.3 (7.7)	2.2 (2.4)	4.4 (4.3)
50–65	17.5 (16.7)	25.0 (26.1)	9.9 (9.6)	16.3 (16.3)
Over 65	72.9 (73.0)	55.6 (55.4)	84.9 (84.9)	73.7 (73.7)

The life-length cdf rises slowly at young ages and sharply at old ages, suggesting the cloglog link (also implied by an exponentially increasing hazard, the Gompertz extreme-value law). For G (1 female) and R (1 black),

$$\log\{-\log[1 - P(Y \le j \mid G = g, R = r)]\} = \alpha_j + \beta_1 g + \beta_2 r, \qquad \hat{\beta}_1 = -0.658,\ \hat{\beta}_2 = 0.626.$$

(Fit statistics are irrelevant: these are population values.) By Proposition 7.12, $P(Y > j \mid G = 0, r) = P(Y > j \mid G = 1, r)^{\exp(0.658)}$: given race, men's survival proportion beyond any age is women's raised to the power 1.93; given gender, blacks' is whites' to the power $e^{0.626} = 1.87$. If white women survive beyond some age with probability π, white men and black women do so with probability about π^2, and black men with about π^4.

7.3.4 Incorporating dispersion effects*

By Proposition 7.7(c) (which holds for any G), cumulative link models force stochastic ordering across predictor settings. If instead responses at $\mathbf{x}_1$ are centred similarly but more dispersed than at $\mathbf{x}_2$, then $P(Y \le j \mid \mathbf{x}_1) > P(Y \le j \mid \mathbf{x}_2)$ for small j but $<$ for large j.

Definition (Location–scale cumulative link model)

$$G^{-1}[P(Y \le j \mid \mathbf{x})] = \frac{\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}}{\exp(\boldsymbol{\gamma}^\top \mathbf{x})}. \tag{7.6}$$

The ordinary model is $\boldsymbol{\gamma} = \mathbf{0}$.

Proposition 7.14 (Two groups with a dispersion effect)

For a dummy x ($x = 1$ for group 1) and the logit link, (7.6) reads $\operatorname{logit} P(Y \le j) = \alpha_j$ for $x = 0$ and $(\alpha_j + \beta)/e^\gamma$ for $x = 1$. The cumulative log odds ratio

$$\operatorname{logit} P(Y \le j \mid x = 1) - \operatorname{logit} P(Y \le j \mid x = 0) = \frac{\alpha_j + \beta}{e^\gamma} - \alpha_j$$

is constant in j iff $\gamma = 0$. If $\gamma > 0$ the group-1 cumulative logits $\alpha_j + \beta$ are shrunk toward 0 (group 1 is more dispersed, with more mass in the end categories); if $\gamma < 0$ they are spread apart (less dispersion).

Proof. The difference equals $\alpha_j(e^{-\gamma} - 1) + \beta e^{-\gamma}$, which is free of j (the α_j being distinct) iff $e^{-\gamma} = 1$. For $\gamma > 0$, $|(\alpha_j + \beta)/e^\gamma| < |\alpha_j + \beta|$, so compared with the pure location shift ($\gamma = 0$) the group-1 cumulative probabilities move toward $\frac{1}{2}$: when $P(Y \le 1) < \frac{1}{2} < P(Y \le J - 1)$, the first increases and the second decreases, putting more probability in categories 1 and J. The case $\gamma < 0$ is the reverse. ■

7.4 Alternative models for ordinal responses*

Motivation. Cumulative logits are not the only way to use ordering. Adjacent-category and continuation-ratio logits describe individual categories or sequential mechanisms, and mean response models mimic ordinary regression.

7.4.1 Adjacent-categories logits

Definition (Adjacent-categories logits)

$$\operatorname{logit} P(Y = j \mid Y \in \{j, j+1\}) = \log \frac{\pi_j}{\pi_{j+1}}, \qquad j = 1, \ldots, J-1. \tag{7.7}$$

Proposition 7.15 (Adjacent-categories and baseline-category logits)

(a) The two sets of logits determine each other:

$$\log \frac{\pi_j}{\pi_J} = \sum_{k=j}^{J-1} \log \frac{\pi_k}{\pi_{k+1}}, \qquad \log \frac{\pi_j}{\pi_{j+1}} = \log \frac{\pi_j}{\pi_J} - \log \frac{\pi_{j+1}}{\pi_J}.$$

(b) The adjacent-categories model with a common effect,

$$\log \frac{\pi_j(\mathbf{x})}{\pi_{j+1}(\mathbf{x})} = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1, \tag{7.8}$$

is the baseline-category logit model $\log[\pi_j(\mathbf{x})/\pi_J(\mathbf{x})] = \alpha_j^* + \boldsymbol{\beta}^\top \mathbf{u}_j$ with $\alpha_j^* = \sum_{k=j}^{J-1} \alpha_k$ and $\mathbf{u}_j = (J - j)\mathbf{x}$.

Proof.

(a) The first identity is a telescoping sum $\sum_{k=j}^{J-1}(\log\pi_k - \log\pi_{k+1}) = \log\pi_j - \log\pi_J$; the second is Proposition 7.1.

(b) Substitute (7.8) into the telescoping sum: $\sum_{k=j}^{J-1}(\alpha_k + \boldsymbol{\beta}^\top\mathbf{x}) = \alpha_j^* + (J-j)\boldsymbol{\beta}^\top\mathbf{x}$. The map is invertible, so the models coincide. ■

So (7.8) can be fitted as a baseline-category model with an adjusted model matrix and one parameter per predictor. Like the proportional odds model it implies stochastic ordering, and the two usually fit well in similar situations. Choose between them by interpretation: adjacent-category effects refer to individual pairs of categories, cumulative effects to the whole scale (hence usually larger, with similar z ratios). Only cumulative logit effects are approximately invariant to the number and choice of categories (Corollary 7.9).

7.4.2 Job satisfaction example

Example 7.16 (Job satisfaction, income and gender)

Black Americans, 1991 General Social Survey. Job satisfaction: very dissatisfied (VD), a little satisfied (LS), moderately satisfied (MS), very satisfied (VS).

Gender	Income ($)	VD	LS	MS	VS
Female	< 5000	1	3	11	2
	5000–15,000	2	3	17	3
	15,000–25,000	0	1	8	5
	> 25,000	0	2	4	2
Male	< 5000	1	1	2	1
	5000–15,000	0	3	5	1
	15,000–25,000	0	0	7	3
	> 25,000	0	1	9	6

With income scores $x = 1,2,3,4$ and $g = 1$ for females, the model $\log(\pi_j/\pi_{j+1}) = \alpha_j + \beta_1 x + \beta_2 g$ is fitted as the baseline-category model $\log(\pi_j/\pi_4) = \alpha_j^* + \beta_1(4-j)x + \beta_2(4-j)g$. ML: $\hat\beta_1 = -0.389$ (SE 0.155), $\hat\beta_2 = 0.045$ (SE 0.314). Given gender, the odds of the lower of two adjacent categories multiply by $e^{-0.389} = 0.68$ per income category. The 24 logits are described by 5 parameters with $G^2 = 12.6$, $df = 19$. The cumulative logit model gives $G^2 = 13.3$ and a larger income effect $\hat\beta_1 = -0.51$ (SE 0.20) since it refers to the whole scale; $\hat\beta_1/\text{SE} \approx -2.5$ in both.

7.4.3 Continuation-ratio logits

Definition (Continuation-ratio logits)

$$\log\frac{\pi_j}{\pi_{j+1}+\cdots+\pi_J} \quad \text{or} \quad \log\frac{\pi_{j+1}}{\pi_1+\cdots+\pi_j}, \qquad j = 1,\ldots,J-1. \tag{7.9}$$

With $\omega_j(\mathbf{x}) = P(Y = j \mid Y \geq j, \mathbf{x}) = \pi_j(\mathbf{x})/[\pi_j(\mathbf{x})+\cdots+\pi_J(\mathbf{x})]$, the first set are the ordinary logits $\operatorname{logit}\omega_j(\mathbf{x})$. They suit sequential mechanisms (survival through successive stages).

Theorem 7.17 (Multinomial factorization into binomials)

Let $(y_{i1},\dots,y_{iJ})$ be multinomial$(n_i;\pi_1,\dots,\pi_J)$ at setting $\mathbf{x}_i$ and $b(n,y;\omega)=\binom{n}{y}\omega^y(1-\omega)^{n-y}$. Then

$$P(y_{i1},\dots,y_{iJ}) = b(n_i,y_{i1};\omega_1)\,b(n_i-y_{i1},y_{i2};\omega_2)\cdots b(n_i-y_{i1}-\cdots-y_{i,J-2},y_{i,J-1};\omega_{J-1}).$$

Consequently, if the models for logit ω_j, $j=1,\dots,J-1$, have distinct parameters, the log likelihood separates into $J-1$ binary-logit log likelihoods: separate fitting equals simultaneous ML fitting, and the separate G^2 statistics add to the overall G^2.

Proof. [Supplement] Write $m_j=n_i-y_{i1}-\cdots-y_{i,j-1}$ (so $m_1=n_i$, $m_J=y_{iJ}$) and $\Pi_j=\pi_j+\cdots+\pi_J$ (so $\Pi_1=1$, $\omega_j=\pi_j/\Pi_j$, $1-\omega_j=\Pi_{j+1}/\Pi_j$).

Step 1. Binomial coefficients: $\prod_{j=1}^{J-1}\binom{m_j}{y_{ij}}=\prod_{j=1}^{J-1}\frac{m_j!}{y_{ij}!\,m_{j+1}!}=\frac{m_1!}{y_{i1}!\cdots y_{i,J-1}!\,m_J!}$, the multinomial coefficient, since the product telescopes and $m_J=y_{iJ}$.

Step 2. Probabilities: $\prod_{j=1}^{J-1}\omega_j^{y_{ij}}(1-\omega_j)^{m_{j+1}}=\prod_{j=1}^{J-1}\frac{\pi_j^{y_{ij}}}{\Pi_j^{y_{ij}}}\cdot\frac{\Pi_{j+1}^{m_{j+1}}}{\Pi_j^{m_{j+1}}}$. Since $y_{ij}+m_{j+1}=m_j$, the jth factor is $\pi_j^{y_{ij}}\Pi_{j+1}^{m_{j+1}}/\Pi_j^{m_j}$, and the product telescopes to $\prod_{j<J}\pi_j^{y_{ij}}\cdot\Pi_J^{m_J}/\Pi_1^{m_1}=\prod_{j=1}^{J}\pi_j^{y_{ij}}$.

Step 3. Multiplying over independent settings i, $\log L=\sum_{j=1}^{J-1}\ell_j(\boldsymbol{\theta}_j)$ where ℓ_j is a binomial log likelihood depending only on the parameters $\boldsymbol{\theta}_j$ of logit ω_j. Maximizing each ℓ_j separately maximizes the sum. The saturated model factorizes in the same way, so $G^2=-2\sum_j(\hat\ell_j-\tilde\ell_j)=\sum_j G_j^2$. ■

Separate fits can use binary logistic software. The analogous statements hold for the second form in (7.9), though it gives different results. Models with common effects across j are sometimes plausible (they then need a joint fit, which is still a binary-logit fit to the stacked data).

7.4.4 Developmental toxicity study with pregnant mice

Example 7.18 (diEGdiME and fetal outcomes)

Pregnant mice exposed to an industrial solvent at five concentrations (mg/kg per day); fetal outcome nonlive, malformed, normal:

Concentration	Nonlive	Malformation	Normal
0 (controls)	15	1	281
62.5	17	0	225
125	22	7	283
250	38	59	202
500	144	132	9

Continuation-ratio models with $x=$ concentration:

$$\log\frac{\pi_1(x)}{\pi_2(x)+\pi_3(x)}=\alpha_1+\beta_1 x,\qquad \log\frac{\pi_2(x)}{\pi_3(x)}=\alpha_2+\beta_2 x.$$

$\hat\beta_1=0.0064$ (SE 0.0004), $\hat\beta_2=0.0174$ (SE 0.0012): the less desirable outcome becomes more likely with concentration. Given a live fetus, the odds of malformation multiply by $e^{1.74}=5.7$ per 100 units. Fit: $G^2=5.78$ and 6.06 ($df=3$ each), total $G^2=11.84$ ($X^2=9.76$, $df=6$, $P=0.07$). Fetuses in a litter are dependent and litters heterogeneous, causing overdispersion; a quasi-likelihood adjustment multiplies SEs by $\sqrt{9.76/6}=1.28$, and both effects remain clearly positive.

7.4.5 Mean response models for ordered response

Definition (Mean response model)

For scores $v_1 \le \cdots \le v_J$ and $M(\mathbf{x}) = \sum_j v_j \pi_j(\mathbf{x})$,

$$M(\mathbf{x}) = \alpha + \boldsymbol{\beta}^\top \mathbf{x}. \tag{7.10}$$

For $J = 2$ it is the linear probability model. It describes only the mean, not the full set of probabilities.

Fitting is by weighted least squares (categorical predictors) or by ML for the product-multinomial likelihood (more complex, since the π_j are not direct functions of the parameters).

7.4.6 Job satisfaction example revisited

Example 7.19 (Job satisfaction: mean response)

With satisfaction and income scores $1, 2, 3, 4$ and gender g, the ML fit is $\hat{M} = 2.59 + 0.181x - 0.030g$ (SE 0.069 for income, 0.145 for gender). Mean satisfaction rises by about 0.2 category per income category; the evidence is strong (Wald $(0.181/0.069)^2 = 6.8$, $P = 0.009$) but the effect is weak: the highest income level averages about $3(0.181) = 0.54$ category above the lowest. WLS gives 0.182 (SE 0.068). Deviance 5.1 with $df = 8 - 3 = 5$.

7.4.7 Advantages and disadvantages of mean response models

Mean response models: pros and cons

(a) *Pro:* they resemble ordinary regression and give simple summaries; for large J they approximate the regression one would fit if Y were measured continuously. Treating an ordinal Y quantitatively is sensible when it crudely measures a continuous variable.
(b) *Con:* fitted means can fall outside $[v_1, v_J]$ (less often for large J and well-dispersed responses); they do not determine cell probabilities, so structural features such as stochastic ordering or independence are not special cases.

7.5 Testing conditional independence in $I \times J \times K$ tables*

Motivation. The CMH test of Section 6.3 extends to multicategory X and Y. Model-based tests and generalized CMH statistics are two faces of the same idea: generalized CMH statistics are (conditional) score tests for multinomial logit models.

7.5.1 Using multinomial models to test conditional independence

Treat Z as a nominal control. The four cases (Y–X types) and their models:

Y–X	Model	Conditional independence	*df*
Ord–Ord	logit $P(Y \le j) = \alpha_j + \beta x_i + \beta_k^Z$	$\beta = 0$	1
Ord–Nom	logit $P(Y \le j) = \alpha_j + \beta_i + \beta_k^Z$	$\beta_1 = \cdots = \beta_I$	$I - 1$
Nom–Ord	$\log[P(Y = j)/P(Y = J)] = \alpha_{jk} + \beta_j x_i$	$\beta_1 = \cdots = \beta_{J-1} = 0$	$J - 1$
Nom–Nom	$\log[P(Y = j)/P(Y = J)] = \alpha_{jk} + \beta_{ij}$	all $\beta_{ij} = 0$	$(I-1)(J-1)$

Here $\{x_i\}$ are ordered scores; the Ord–Ord model is

$$\operatorname{logit} P(Y \le j \mid X = i, Z = k) = \alpha_j + \beta x_i + \beta_k^Z, \tag{7.11}$$

and the conditional independence model for nominal Y is $\log[P(Y = j \mid X = i, Z = k)/P(Y = J \mid X = i, Z = k)] = \alpha_{jk}$. These tests work well when the model captures a major component of the departure from independence, which need not be checked by a fit test first. Allowing effects to vary by stratum multiplies df by K and costs power; with ordinal Z, a log odds ratio changing linearly in Z is an intermediate option.

7.5.2 Job satisfaction example revisited

Example 7.20 (Job satisfaction: model-based tests)

Income scores $3, 10, 20, 35$ (thousands of dollars), controlling for gender. LR statistics against the model without income:

Satisfaction	Income	G^2 fit	df	Test	df	P
Ordinal	Ordinal	13.95	19	5.7	1	0.017
	Nominal	10.51	17	9.1	3	0.028
	Not in model	19.62	20	–	–	–
Nominal	Ordinal	11.74	15	7.6	3	0.054
	Nominal	7.09	9	12.3	9	0.198
	Not in model	19.37	18	–	–	–

The $df = 1$ trend test gives the strongest evidence; nominal treatments dilute it.

7.5.3 Generalized Cochran–Mantel–Haenszel tests for $I \times J \times K$ tables

Given the row and column totals of stratum k, the table has $(I-1)(J-1)$ free counts $\mathbf{n}_k = (n_{11k}, \ldots, n_{1,J-1,k}, \ldots, n_{I-1,J-1,k})^\top$. Under conditional independence the stratum is multiple hypergeometric with $\mathbf{m}_k = E(\mathbf{n}_k)$, $m_{ijk} = n_{i+k}n_{+jk}/n_{++k}$, and

$$\operatorname{Cov}(n_{ijk}, n_{i'j'k}) = \frac{n_{i+k}(\delta_{ii'}n_{++k} - n_{i'+k})\, n_{+jk}(\delta_{jj'}n_{++k} - n_{+j'k})}{n_{++k}^2(n_{++k}-1)},$$

the matrix V_k. Let $\mathbf{n} = \sum_k \mathbf{n}_k$, $\mathbf{m} = \sum_k \mathbf{m}_k$, $V = \sum_k V_k$.

Definition (Generalized CMH statistics)

(a) Nominal X and Y (general association):

$$\text{CMH} = (\mathbf{n} - \mathbf{m})^\top V^{-1} (\mathbf{n} - \mathbf{m}), \qquad df = (I-1)(J-1). \tag{7.12}$$

(b) Ordinal X and Y with scores $\{u_i\}, \{v_j\}$ (Mantel's nonzero correlation statistic): with $T_k = \sum_i \sum_j u_i v_j n_{ijk}$,

$$M^2 = \frac{\left[\sum_k \{T_k - E(T_k)\}\right]^2}{\sum_k \operatorname{Var}(T_k)}, \qquad df = 1, \tag{7.13}$$

where $E(T_k) = (\sum_i u_i n_{i+k})(\sum_j v_j n_{+jk})/n_{++k}$ and

$$\operatorname{Var}(T_k) = \frac{1}{n_{++k}-1}\Big[\sum_i u_i^2 n_{i+k} - \frac{(\sum_i u_i n_{i+k})^2}{n_{++k}}\Big]\Big[\sum_j v_j^2 n_{+jk} - \frac{(\sum_j v_j n_{+jk})^2}{n_{++k}}\Big].$$

(c) Nominal X, ordinal Y (row mean scores differ): a statistic of the Landis et al. family comparing the I row mean scores to their null expectations, $df = I - 1$.

For one stratum, $[T - E(T)]/\sqrt{\text{Var}(T)} = \sqrt{n-1}\,r$, so $M^2 = (n-1)r^2$ is the trend statistic of Chapter 3.

Proposition 7.21 (One stratum: CMH is a rescaled Pearson statistic)

For $K = 1$ and a table with n observations, CMH $= \frac{n-1}{n}X^2$, where X^2 is Pearson's statistic for independence.

Proof. [Supplement] Let $r_i = n_{i+}/n$, $c_j = n_{+j}/n$, and $A = \text{diag}(r_1, \ldots, r_{I-1}) - \mathbf{r}\mathbf{r}^\top$, $B = \text{diag}(c_1, \ldots, c_{J-1}) - \mathbf{c}\mathbf{c}^\top$ (with $\mathbf{r} = (r_1, \ldots, r_{I-1})^\top$, $\mathbf{c}$ similarly).

Step 1. Since $n_{i+}(\delta_{ii'}n - n_{i'+}) = n^2(\delta_{ii'}r_i - r_i r_{i'})$, the covariance formula gives $V = \frac{n^2}{n-1}A \otimes B$.

Step 2. One checks directly that $A^{-1} = \text{diag}(1/r_i) + \mathbf{1}\mathbf{1}^\top/r_I$: indeed $(\text{diag}(r) - \mathbf{r}\mathbf{r}^\top)(\text{diag}(1/r) + \mathbf{1}\mathbf{1}^\top/r_I) = I + \mathbf{r}\mathbf{1}^\top/r_I - \mathbf{r}\mathbf{1}^\top - \mathbf{r}(1-r_I)\mathbf{1}^\top/r_I = I$, using $\mathbf{1}^\top\mathbf{r} = 1 - r_I$. Similarly for B^{-1}.

Step 3. Let $d_{ij} = n_{ij} - m_{ij}$ for all cells; rows and columns of (d_{ij}) sum to 0. Then

$$\mathbf{d}^\top(A^{-1} \otimes B^{-1})\mathbf{d} = \sum_{i,i'<I}\sum_{j,j'<J} d_{ij}d_{i'j'}\Big(\frac{\delta_{ii'}}{r_i} + \frac{1}{r_I}\Big)\Big(\frac{\delta_{jj'}}{c_j} + \frac{1}{c_J}\Big)$$

$$= \sum_{i<I,j<J}\frac{d_{ij}^2}{r_i c_j} + \sum_{i<I}\frac{(\sum_{j<J} d_{ij})^2}{r_i c_J} + \sum_{j<J}\frac{(\sum_{i<I} d_{ij})^2}{r_I c_j} + \frac{(\sum_{i<I,j<J} d_{ij})^2}{r_I c_J}.$$

Step 4. The zero margins give $\sum_{j<J} d_{ij} = -d_{iJ}$, $\sum_{i<I} d_{ij} = -d_{Ij}$ and $\sum_{i<I,j<J} d_{ij} = d_{IJ}$, so the four sums combine into $\sum_{\text{all } i,j} d_{ij}^2/(r_i c_j)$. Since $m_{ij} = n r_i c_j$, this is $n\sum_{ij} d_{ij}^2/m_{ij} = nX^2$.

Step 5. Hence CMH $= \frac{n-1}{n^2}\cdot nX^2 = \frac{n-1}{n}X^2$. ■

7.5.4 Job satisfaction example revisited

Example 7.22 (Job satisfaction: generalized CMH)

With income scores $3, 10, 20, 35$ and satisfaction scores $1, 3, 4, 5$, controlling for gender:

Alternative	*df*	Value	*P*
Nonzero correlation (M^2)	1	6.156	0.013
Row mean scores differ	3	9.034	0.029
General association (7.12)	9	10.200	0.335

As with the model-based tests, the ordinal $df = 1$ test gives the strongest evidence.

7.5.5 Related score tests for multinomial logit models

Theorem 7.23 (Generalized CMH statistics are conditional score tests)

Consider independent multinomial rows in each stratum with $\log[P(Y = j \mid X = i, Z = k)/P(Y = J \mid X = i, Z = k)] = \alpha_{jk} + \tau_{ij}$, $\tau_{Ij} = \tau_{iJ} = 0$ (model (7.14) below), and condition on all row and column totals in every stratum.
(a) The conditional score statistic for H_0: all $\tau_{ij} = 0$ is the generalized CMH statistic (7.12).
(b) For the submodel $\tau_{ij} = \beta u_i(v_j - v_J)$ the conditional score statistic for $H_0 : \beta = 0$ is M^2 (7.13) with the same scores. With $v_j = -j$ this submodel is the adjacent-categories model $\log(\pi_j/\pi_{j+1}) = \alpha'_{jk} + \beta u_i$, so M^2 with equally spaced column scores is its score test.

The nominal–nominal model referred to is

$$\log \frac{P(Y=j \mid X=i, Z=k)}{P(Y=J \mid X=i, Z=k)} = \alpha_{jk} + \tau_{ij}. \tag{7.14}$$

Proof. [Supplement]

Step 1. In stratum k the rows are multinomial$(n_{i+k}; \pi_{1|ik}, \ldots, \pi_{J|ik})$ with $\pi_{j|ik} = \pi_{J|ik}\exp(\alpha_{jk}+\tau_{ij})$. The likelihood is

$$\prod_k \prod_i \frac{n_{i+k}!}{\prod_j n_{ijk}!} \pi_{J|ik}^{n_{i+k}} \exp\Big(\sum_j \alpha_{jk} n_{ijk} + \sum_j \tau_{ij} n_{ijk}\Big),$$

so the sufficient statistics are n_{+jk} for α_{jk} and $n_{ij+} = \sum_k n_{ijk}$ for τ_{ij} (the row totals are fixed).

Step 2. Conditioning on $\{n_{+jk}\}$ removes the α_{jk} and the factors $\pi_{J|ik}^{n_{i+k}}$, which depend only on fixed totals:

$$P(\{n_{ijk}\} \mid \text{margins}) = \frac{\prod_k \prod_{ij} (n_{ijk}!)^{-1} \exp(\sum_{ij} \tau_{ij} n_{ij+})}{\sum_{\text{tables with these margins}} \prod_k \prod_{ij} (n^*_{ijk}!)^{-1} \exp(\sum_{ij} \tau_{ij} n^*_{ij+})}.$$

This is an exponential family in $\boldsymbol{\tau} = (\tau_{ij})_{i<I, j<J}$ with sufficient statistic $\mathbf{n} = (n_{ij+})_{i<I, j<J}$.

Step 3. For such a family the score is $\mathbf{n} - E_{\boldsymbol{\tau}}(\mathbf{n})$ and the information is $\mathrm{Cov}_{\boldsymbol{\tau}}(\mathbf{n})$. At $\boldsymbol{\tau} = \mathbf{0}$ the strata are independent multiple hypergeometric, so $E_0(\mathbf{n}) = \mathbf{m}$ and $\mathrm{Cov}_0(\mathbf{n}) = V$. The score statistic $(\mathbf{n}-\mathbf{m})^\top V^{-1}(\mathbf{n}-\mathbf{m})$ is (7.12), proving (a).

Step 4. For $\tau_{ij} = \beta u_i (v_j - v_J)$, $\sum_{ij} \tau_{ij} n_{ij+} = \beta \sum_k \sum_{ij} u_i (v_j - v_J) n_{ijk} = \beta(\sum_k T_k - v_J \sum_k \sum_i u_i n_{i+k})$. The second term is fixed by the margins, so the conditional family has sufficient statistic $\sum_k T_k$; its score at $\beta = 0$ is $\sum_k [T_k - E_0(T_k)]$ and its information is $\sum_k \mathrm{Var}_0(T_k)$ (strata independent). Their ratio is M^2.

Step 5. With $v_j = -j$, $\tau_{ij} = \beta u_i (J - j)$, which by Proposition 7.15(b) is the baseline form of $\log(\pi_j/\pi_{j+1}) = \alpha'_{jk} + \beta u_i$. ∎

Other score-test equivalences

M^2 with average-rank (midrank) column scores is the score test of $\beta = 0$ in the cumulative logit model (7.11) (McCullagh); for $2 \times J$ tables this is a discrete Wilcoxon–Mann–Whitney test. With large strata, generalized CMH and LR tests agree closely; the model-based approach gives effect estimates, while the CMH tests remain valid under sparse asymptotics ($K \to \infty$).

7.5.6 Exact tests of conditional independence

Conditioning on row and column totals in each stratum, the null distribution of the counts is a product of multiple hypergeometric distributions, which induces an exact distribution for any generalized CMH statistic. The exact P-value is the probability of tables with the observed strata margins whose statistic is at least as large as observed.

7.6 Discrete-choice multinomial logit models*

Motivation. In choice problems (travel mode, brand, shopping destination) some predictors describe the options rather than the chooser, and the set of options may vary by subject. McFadden's conditional logit model handles both, and contains the baseline-category model as a special case.

7.6.1 Discrete-choice modeling

Predictors may be *characteristics of the choices* (cost and time of each travel mode, price of each brand), which vary across options, or *characteristics of the chooser* (income, education), constant across options.

Definition (McFadden's conditional logit model)

For subject i with choice set C_i and option-specific predictor vectors $\mathbf{x}_{ij}$,

$$\pi_j(\mathbf{x}_i) = \frac{\exp(\boldsymbol{\beta}^\top \mathbf{x}_{ij})}{\sum_{h \in C_i} \exp(\boldsymbol{\beta}^\top \mathbf{x}_{ih})}, \qquad j \in C_i. \tag{7.15}$$

Proposition 7.24 (Pairwise logits and independence from irrelevant alternatives)

Under (7.15), for $a, b \in C_i$,

$$\log \frac{\pi_a(\mathbf{x}_i)}{\pi_b(\mathbf{x}_i)} = \boldsymbol{\beta}^\top (\mathbf{x}_{ia} - \mathbf{x}_{ib}). \tag{7.16}$$

The odds of a over b depend neither on the other options in C_i nor on their predictor values.

Proof. The denominators in (7.15) are common to π_a and π_b, so $\pi_a/\pi_b = \exp(\boldsymbol{\beta}^\top \mathbf{x}_{ia})/\exp(\boldsymbol{\beta}^\top \mathbf{x}_{ib})$, which involves only options a and b. ■

A variable with equal values for a and b has no influence on the choice between them. The IIA property can be unrealistic: if 80% choose auto over a red bus (odds 4), adding an identical blue bus should leave about 10% for each bus (odds 8 of auto vs. red bus), but (7.16) keeps the odds at 4. The model is appropriate when the options are distinct and weighed independently.

7.6.2 Discrete-choice and multinomial logit models

Proposition 7.25 (Baseline-category logits as a discrete-choice model)

For a chooser characteristic x_i, let $\delta_{jk} = \mathbb{1}\{j = k\}$, and define for option j

$$\mathbf{z}_{ij} = (\delta_{j1}, \ldots, \delta_{jJ},\ \delta_{j1}x_i, \ldots, \delta_{jJ}x_i)^\top, \qquad \boldsymbol{\beta} = (\alpha_1, \ldots, \alpha_J, \beta_1, \ldots, \beta_J)^\top.$$

Then $\boldsymbol{\beta}^\top \mathbf{z}_{ij} = \alpha_j + \beta_j x_i$, and the discrete-choice model (7.15) in $\mathbf{z}_{ij}$ with $\alpha_J = \beta_J = 0$ is exactly the baseline-category model (7.2).

Proof. Only the jth dummy and the jth product are nonzero in $\mathbf{z}_{ij}$, so $\boldsymbol{\beta}^\top \mathbf{z}_{ij} = \alpha_j + \beta_j x_i$. Then

$$\frac{\exp(\boldsymbol{\beta}^\top \mathbf{z}_{ij})}{\sum_{h=1}^{J} \exp(\boldsymbol{\beta}^\top \mathbf{z}_{ih})} = \frac{\exp(\alpha_j + \beta_j x_i)}{\sum_{h=1}^{J} \exp(\alpha_h + \beta_h x_i)},$$

which with $\alpha_J = \beta_J = 0$ is (7.2). ■

Thus (7.15) can mix characteristics of choices and choosers, and is very general.

7.6.3 Shopping choice example

Example 7.26 (Shopping destinations in Pittsburgh)

Residents chose among five city zones. Predictors: shopping opportunity S (zone's share of regional retail employment) and trip price P (from auto travel time and operating cost). Estimates: -1.06 (SE 0.28) for price, 0.84 (SE 0.23) for opportunity, so by (7.16)

$$\log \frac{\hat{\pi}_a}{\hat{\pi}_b} = -1.06(P_a - P_b) + 0.84(S_a - S_b).$$

A destination becomes more attractive as its trip price falls and its shopping opportunity rises; (7.15) gives the estimated probability of each destination.

7.7 R lab

Motivation. Fit the chapter's main models in R: baseline-category logits (`nnet::multinom`), proportional odds (`MASS::polr`), continuation-ratio logits as separate binary fits, and generalized CMH tests.

Baseline-category logits (alligators, collapsed over gender). A matrix response of counts is allowed; the first column (fish) is the baseline. Coefficients reproduce the alligator estimates table.

```
library(nnet)
gator <- data.frame(lake = rep(c("Hancock","Oklawaha","Trafford","George"),
                               each = 2), size = rep(c("small","large"), 4))
gator$Y <- matrix(c(23,4,2,2,8, 7,0,1,3,5, 5,11,1,0,3, 13,8,6,1,0,
                    5,11,2,1,5, 8,7,6,3,5, 16,19,1,2,3, 17,1,0,1,3),
                  ncol = 5, byrow = TRUE, dimnames = list(NULL,
                  c("Fish","Invert","Rept","Bird","Other")))
gator$lake <- relevel(factor(gator$lake), ref = "George")
gator$size <- relevel(factor(gator$size), ref = "large")
fit <- multinom(Y ~ size + lake, data = gator)
coef(fit); summary(fit)$standard.errors       # Invert: -1.55, 1.46, ...
predict(fit, data.frame(size = "large", lake = "Hancock"), type = "probs")
```

Proportional odds (mental impairment). `polr` uses logit $P(Y \le j) = \zeta_j - \boldsymbol{\beta}^\top \mathbf{x}$, so its slopes are the negatives of those in the text (0.319, -1.111). `method="probit"` or `"cloglog"` gives other cumulative links.

```
library(MASS)
mental <- data.frame(impair = rep(1:4, c(12, 12, 7, 9)),
  ses  = c(1,1,1,1,0,1,0,1,1,1,0,0, 1,0,1,0,1,1,0,1,1,0,1,1,
           0,1,0,0,1,0,0, 1,1,1,0,0,0,1,0,0),
  life = c(1,9,4,3,2,0,1,3,3,7,1,2, 5,6,3,1,8,2,5,5,9,3,3,1,
           0,4,3,9,6,4,3, 8,2,7,5,4,4,8,8,9))
pfit <- polr(factor(impair) ~ life + ses, data = mental)
summary(pfit)                                  # zeta: -0.282, 1.213, 2.209
predict(pfit, data.frame(life = 4.275, ses = c(0, 1)), type = "probs")
polr(factor(impair) ~ life + ses, data = mental, method = "probit")
```

Continuation-ratio logits (pregnant mice). By Theorem 7.17 the two binary fits are the joint ML fit, and their deviances add.

```
conc <- c(0, 62.5, 125, 250, 500)
nonlive <- c(15,17,22,38,144); malf <- c(1,0,7,59,132)
normal  <- c(281,225,283,202,9)
```

```
f1 <- glm(cbind(nonlive, malf + normal) ~ conc, family = binomial)
f2 <- glm(cbind(malf, normal) ~ conc, family = binomial)
rbind(coef(f1), coef(f2))                    # slopes 0.0064, 0.0174
deviance(f1) + deviance(f2)                  # 11.84 on df = 6
```

Generalized CMH tests (job satisfaction). `mantelhaen.test` on an $I \times J \times K$ array gives the general-association statistic (7.12); a few lines compute Mantel's M^2 (7.13).

```
job <- array(c(1,2,0,0, 3,3,1,2, 11,17,8,4, 2,3,5,2,
               1,0,0,0, 1,3,0,1, 2,5,7,9, 1,1,3,6), dim = c(4,4,2),
  dimnames = list(income = 1:4, satis = 1:4, gender = c("F","M")))
mantelhaen.test(job)                         # 10.20, df = 9
M2 <- function(tab, u, v) {
  num <- 0; den <- 0
  for (k in seq_len(dim(tab)[3])) {
    t <- tab[, , k]; n <- sum(t); r <- rowSums(t); cc <- colSums(t)
    num <- num + sum(outer(u, v) * t) - sum(u * r) * sum(v * cc) / n
    den <- den + (sum(u^2 * r) - sum(u * r)^2 / n) *
                 (sum(v^2 * cc) - sum(v * cc)^2 / n) / (n - 1)
  }
  num^2 / den
}
M2(job, u = c(3, 10, 20, 35), v = c(1, 3, 4, 5))   # 6.156, df = 1
```